MIL-F-43782A
5 June 1981
SUPERSEDING
MIL-F-43782
30 May 1972

MILITARY SPECIFICATION

FOOT MEASURING DEVICE, MEN'S

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Score. This specification covers a foot measuring device for men.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-225/8	-	Aluminum Alloy Bar, Rod, Wire and Special Shapes; Rolled, Drawn, or Cold Finished, 6061
QQ-A-250/11	-	Aluminum Alloy 6061, Plate and Sheet
TT-L-58	-	Lacquer; Spraying, Clear and Pigmented, (General Interior Use)
TT-P-1757	-	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8335

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:20Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

2-1-1095 - Foot Measuring Device; Assembly Complete
2-1-1096 - Foot Measuring Device; Left and Right Foot Measuring Device Assemblies
2-1-1097 - Foot Measuring Device; Plate Detail
2-1-1098 - Foot Measuring Device; Miscellaneous Details
2-1-1099 - Foot Measuring Device; Base Detail
2-1-1100 - Foot Measuring Device; Scales and Label
2-1-1101 - Foot Measuring Device; Detail of Label Placement

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS (NBS)

PS1 - Softwood Plywood/Construction and Industrial

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Alternate components. Components offered as equivalent to components specified hereinafter and on the reference drawings as a specific manufacturer's part number or equal shall be functionally equal to the manufacturer's part number

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:20Z
Check the source to verify that this is the current version before use.

identified and of equal or better quality. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the first article or if none is required, prior to commencing production, the contractor shall submit for the contracting officers' approval, a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of the first article or production manufacture, as applicable, and does not relieve the contractor of responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.3).

3.3 Materials and components. The materials and components shall be as specified herein and as shown on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this specification (see 6.5).

3.3.1 Aluminum alloy. Aluminum alloy bar, rod and wire shall conform to temper T-6 of QQ-A-225/8. Aluminum alloy plate and sheet shall conform to temper T-6 of QQ-A-250/11.

3.4 Design and construction. The design and construction of the foot measuring device shall be as specified herein, and as shown on the drawings referenced in 2.1. The wood base shall be five ply plywood, group B, grade AB conforming to PS1.

3.5 Manufacturers label. The manufacturers label shall be a plate or photo-sensitive metal etched, stamped, processed, or painted, using characters 3/16 inch high, with the name of the manufacturer and year of manufacture. The label shall be fastened securely to the base of the device.

3.6 Finish. Aluminum components (except the scales and labels) shall be coated with a primer conforming to TT-P-1757 followed by a coat of black lacquer conforming to type II, class 1 or 2 of TT-L-58. Scales and labels shall be coated with clear lacquer conforming to type 1, class 1 of TT-L-58 after installation. The plywood bases shall be coated with a clear high gloss varnish that will produce a uniform finish.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:20Z
Check the source to verify that this is the current version before use.

3.7 Government furnished property. The Government shall furnish gauge blocks consisting of size 9R right and left foot patterns (see 6.2 and 6.4). The gauge blocks are intended to serve as a convenient means of verifying scale placement and shall not be used as an alternate method for locating the scales.

3.8 Workmanship. The finished foot measuring device shall be free from fractures, splits, punctures, tears, dents, creases, bows, miscasts, deteriorations or malfunctions. There shall be no sharp edges, burrs, or slivers. The finish shall be free of smudges, poor touch up, scale, scratches, stain, overspray, sag, imprint or embedded foreign matter. All bolts, studs, nuts and screws shall be drawn tight. The surface of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be sound, continuous, smooth and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed from the finished welds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3.1 and 4.4.3.2 and tested as specified in 4.5.1. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified, or qualified in this specification or aplicable purchase document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:20Z
Check the source to verify that this is the current version before use.

4.4.2 In-process inspection. Examination shall be made during the manufacturing process for priming specified in 3.6. Whenever nonconformance is noted, correction shall be made to the applicable process and all items processed.

4.4.3 End item inspection. The end item shall be inspected in accordance with 4.4.3.1, 4.4.3.2 and 4.4.3.3. The lot size shall be expressed in units of one foot measuring device.

4.4.3.1 Visual examination. The end item shall be examined for defects listed in table I. The inspection level shall be II and the acceptable quality level (AQL) shall be 2.5 major and 6.5 total (major and minor combined) defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Color not as specified	X	
	Poor touch-up, smudge, scale, scratched, stained, overspray, sag, imprint or foreign matter imbedded		X
Design, construction and workmanship, general (applicable to all components and assemblies)	Any component or part missing or not specified type or size,	X	
	sharp edge, burr or sliver	X	
	Fractured, split, punctured, torn, dented, creased, deteriorated or malformed	X	
Welding	Missing, incomplete, burn holes, cracked, fractured, or not fused	X	
	Undercut, not smooth and uniform, or scale or flux deposits not removed		X
Bolts, nuts, screws, studs and other type of threaded fasteners	Broken, stripped, fractured, or not tight		X
Markings, scales, and labels	Missing, incomplete, or not legible	X	

Source: https://assist.dla.mil -- Downloaded: 202640-21T22:20Z
Check the source to verify that this is the currenversion before use.

4.4.3.2 Dimensional examination. The end item shall be examined for compliance with dimensions specified. Any noncompliance therewith shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

4.4.3.3 End item testing. The completely finished measuring device shall be tested as specified in 4.5.1 and 4.5.2. The inspection level shall be S-2 and any failure of sample unit shall be cause for rejection of the lot.

4.4.4 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with table II. The sample unit shall be one shipping container fully packaged except that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully packaged. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

TABLE II. Packaging defects

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping or tape banding, and inadequate stapling.
	Bulged or distorted container.
Preservation	Lubrication missing.
Content	Number of unit packs per container is more or less than required.

4.5 Methods of inspection.

4.5.1 Operational test for calibration reliability. The slides shall be pulled forward and back, fully opening and closing the measuring area for at least six times. The alignment of the length and width scales shall then be checked for conformance with drawing 2-1-1101.

4.5.2 Check test for size and width. The size 9R gage blocks (GFP) shall be used to check the correct position for the attachment of the size and width scales.

Source: https://assist.dla.mil -- Downloaded: 202640-21T22:20Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each device shall be thoroughly cleaned and lubricated, with the manufacturer's recommended lubricant. Each device shall be unit packed in a fiberboard box conforming to style FTC, type CF, variety optional, class domestic, grade 125 of PPP-B-636. Each box shall be provided with two corrugated fiberboard top pads conforming to type CF, class domestic, variety SW, grade 275 of PPP-F-320 and measuring 1/4 inch less than the inside length and width dimensions of the box, placed directly over the device. Box closure shall be in accordance with method II of the appendix of PPP-B-636.

5.1.2 Commercial. Devices shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Five devices, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.2 Level B packing. Five devices, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification, except that the inspection shall be in accordance with 4.4.4.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box, fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the box specification, except that the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Devices, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract, unit packs and shipping containers shall be marked in accordance with MIL-STD-1188, as applicable.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:20Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The foot measuring device is intended for use by induction centers and supply stations of the Department of Defense to determine footwear sizes for men.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.2, 4.2 and 6.3).
(c) Source and delivery of Government furnished property (see 3.8 and 6.4).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) Whether weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the DAR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished.

6.4 Government furnished property. The contracting officer should arrange to furnish the property specified in 3.7 and 6.2.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Air Force - 11

Review activity:

Navy - NU
Air Force - 99
DLA - CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

Project No. 8335-0120

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:20Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and *enhance its use*. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-43782A Foot Measuring Device, Men's

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. *REMARKS*

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 20260-21T22:20Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development
Laboratories
ATTN: DRDNA-ES
Natick, MA 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:20Z
Check the source to verify that this is the current version before use.